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CMPT 353 - Project Report

Pokemon Trading Card Business Analysis

Introduction

I wanted to do a data science project related to business but I ran into a problem. I had to figure out a way to collect real world transactions without owning a real business myself. My original idea was to gather data for a clothing company since I had interest in starting one myself. However, collecting data from existing companies posed a problem as most of the information was confidential. The next idea I had was using my transactions of selling and buying Pokemon cards. I was in the process of selling my personal collection to move on from the hobby but thought the information from selling reflects the business inspiration I had for this project. Currently Pokemon cards pose a high interest in consumers (Philip Collectibles, 2026; TwiceBakedJake, 2026). This has caught the eyes of investors as interest gain surpasses the stock market with cards increasing by 3000% in value (Sullivan, 2025). Creating a model to analyze the business of Pokemon cards as well as predict which cards will out perform others will pose to be useful as the bull market continues. Pokemon cards are at peak interest from nostalgic collectors and newcomers who enjoy artwork. The demand for them will seemingly increase as more highly anticipated Pokemon sets (of cards) are released.

Questions

The questions I want to answer with my data analysis project are:

- Is investing in Pokemon cards viable?
- Is starting a business selling cards viable?
- What are the risks in this business?
- Which products are the best to continue the business?
- Which products are best in day to day sales?

Most of which will be answered through statistical analysis tools learned throughout the class. As well as personal research to better know the market and its behaviour (which will be discussed during the analysis report).

How I received the data

While working at a card show, I started logging my personal collection sales through a discord server, which is a common way most sellers log their transactions. When a sale is made, I take a picture, number the amount and send it to a discord server tab (seen in figure 1). That way all the transactions are recorded with the timestamp that is conveniently built in from discord. I take all the logs and manually add them in a Google spreadsheet before exporting the data to a .csv file (as seen in figure 2). Most, if not, all of the data here has been altered slightly to keep confidentiality but the impact of the data remains unaffected. Sales, profits or timing will be displayed differently but still reflects real world transactions and will not impact the model in any way.

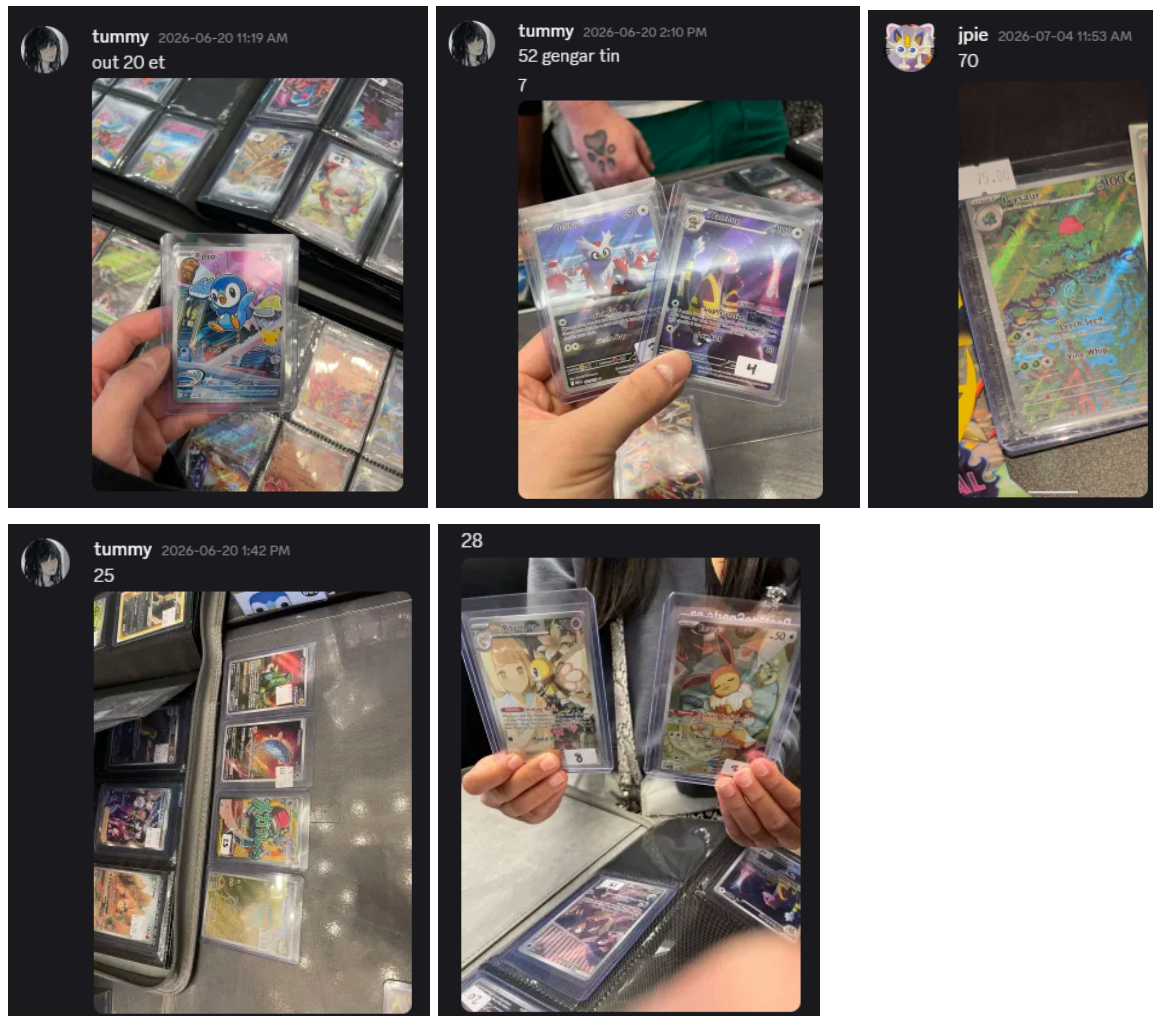


Figure 1: Some of the discord logs during one of my card shows

1	item	product type	set	cost	sold	show	show time	purchased date	sold date	notes
2	piplup	card	first partner	6.67	20	guildford card sh	11:19	2026-03-20	2026-06-20	
3	feraligatr ex box	box	ascended heroes	50	70	guildford card sh	11:26		2026-06-20	
4	nintails	card	obsidian flames	30	60	guildford card sh	12:32		2026-06-20	
5	ducklett jp	card	black white	5	15	guildford card sh	12:40		2026-06-20	
6	golurk jp	card	black white	3	5	guildford card sh	12:40		2026-06-20	
7	zerora vmax	card	crown zenith	5	20	guildford card sh	12:55		2026-06-20	

Figure 2: Logs written manually in a Google spreadsheet (a portion of the data)

I found my transactions were not enough to make an analysis on (or at least enough I was satisfied with for the project). So I ended up gathering logs from colleagues who also work at the card shows (events where you can sell cards). Their businesses are much larger than my personal selling so their logs better represent the current market. As well, these logs are recorded the same way through discord. I just needed to manually input them in Google sheets and export to .csv. These logs have also been altered slightly to hold confidentiality and anomaly but still accurately reflects the market and does not impact the model.

Cleaning the data

Of course there will be missing data values in some categories in cases of transaction errors. Such as forgetting when an item was sold, an item's purchase time or the cost amount of an item. The script [01-clean-missing-values.py](#) handles these situations by setting default values. For cost the default is 70% of the sold price as the regular rate to buy an item (if one was a seller at these card events) is 70% of its market value. The time is defaulted to 12:00pm of the sold date and the purchase date is defaulted to April 1st, 2026 (the start of the semester) for the sake of the investment portion of the business analysis. Lastly, the script handles better formatting by combining the sold date and time to make a datetime object for the system during analysis.

Sales Analysis

Stats for show: All Sales	Stats for show: guildford_card_show
Tax Rate: 0.12	Tax Rate: 0.12
Total Revenue: \$6930.00	Total Revenue: \$1702.00
Total Taxes: \$831.60	Total Taxes: \$204.24
Total Profits: \$2095.93	Total Profits: \$412.79
Total Profit Margin: 0.302	Total Profit Margin: 0.243
Average Selling Price: \$70	Average Selling Price: \$43.6
Median Selling Price: \$25	Median Selling Price: \$23
Average Profit per Item: \$21.2	Average Profit per Item: \$10.6
Peak of Sale times: 13:00	Peak of Sale times: 13:00
Amount of Sales: 99	Amount of Sales: 39
Amount of Sales Over \$100: 16	Amount of Sales Over \$100: 4
Amount of Sales Between \$50-\$99: 15	Amount of Sales Between \$50-\$99: 7
Amount of Sales Below \$50: 68	Amount of Sales Below \$50: 28
Set with Highest Sales: crown zenith	Set with Highest Sales: neo genesis
Total Profit from Highest Set: \$965	Total Profit from Highest Set: \$109
Type with Highest Sales: card	Type with Highest Sales: card
Amount of Card Sales: 87	Amount of Card Sales: 35
Amount of Sealed Sales: 5	Amount of Sealed Sales: 4
Amount of Graded Card Sales: 7	Amount of Graded Card Sales: 0
Total Card Profits: \$1344.67	Total Card Profits: \$394.03
Total Sealed Profits: \$86.26	Total Sealed Profits: \$18.76
Total Graded Card Profits: \$665.00	Total Graded Card Profits: \$0.00
Average Card Profit: \$15.5	Average Card Profit: \$11.3
Average Sealed Profit: \$17.3	Average Sealed Profit: \$4.69
Average Graded Card Profit: \$95	Average Graded Card Profit: \$nan

Figure 3: Sales stats for all sales and one card show (Guildford Card Show)

Running the [03-analyze-sales.py](#) script from the system prints out sales stats shown in figure 3. As we can see, the stats give us useful information. Looking at the Total Profits of \$2095.93 and Total Profit Margin of 30%, this shows the business is extremely viable. Most businesses strive for a 10-15% profit margin or more as that indicates a more financially successful business (Brex, n.d.; Hanna, 2026). These results from three card shows, a few outside sales and 99 overall sales already show promising potential. Additionally, looking at the inventory metrics, the most successful set of cards is from Crown Zenith. This set was known to be a fan favorite and is considered one of the best sets ever printed from the Pokemon Company. The stats indicate that buys from Crown Zenith should be prioritized as it is the highest selling set from our analysis.

Investment Analysis

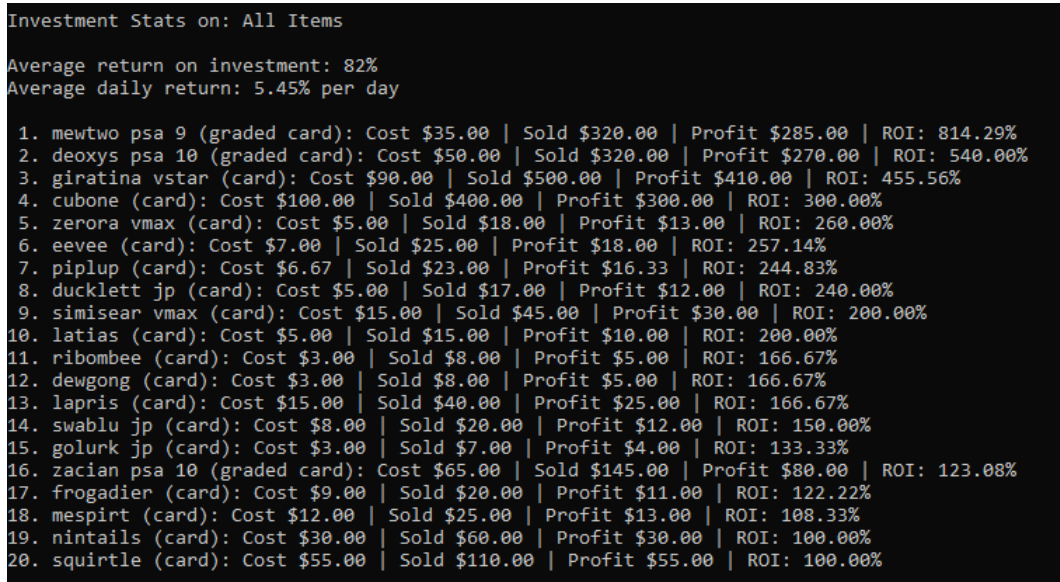


Figure 4: Investment stats for all sales and top 20 items with highest ROI.

The first statistic to look at in figure 4 is the 82% average return on investment. 82% is considered an exceptionally high result as most investments average an 8-10% return (Heath, 2020). Looking at the bull market of Pokemon cards, this does not seem surprising but is still an astonishing high number for a return rate. Even the top twenty items with highest return on investment (ROI) show incredible numbers. This indicates that investment in Pokemon cards is actually as viable, if not, more viable than starting a card business.

Sales Plot

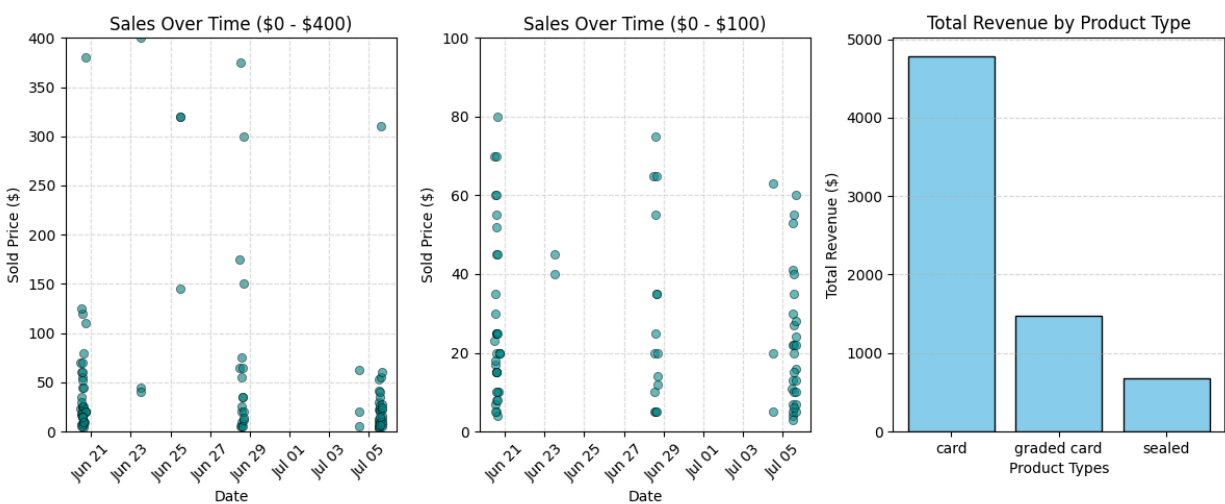


Figure 5: All sales plotted over time. Bar graph of revenue by product types.

The sales plot generated by the [05-generate-plots-all-sales.py](#) script shows all the sales by date as well as all revenue from each product type (figure 5). The purpose of the plots is to display business statistics for easy readability and analysis. As we can see, most of the sales are from single cards that are below \$50. From a business standpoint, card vendors should prioritize lower value single cards over higher value ones for better results and faster sales. Of course this can change as time goes on so gathering more data points and sales will help analyze the right decisions. For additional plots and a more detailed look of each individual show refer to figure 6, 7 and 8.

Limitations and Future Considerations

Ninety-nine data points in three shows were collected over the one month duration of the project. If the experiment were conducted over a whole year instead, there would be a significant difference in business statistics over the analysis we received from the project. Due to the nature of the model being a school project with a few weeks to experiment with, sales analysis of three shows was the most I could gather in the allotted time. If I were to do this project again I would love to run the system and gather data over the course of a year.

Some scripts had to be manually coded to account for naming of product types and input files. Product types follow three categories - cards, graded cards and sealed. If a graded card was inputted as slab (ambiguous terms) there could be two different categories representing the same product type. Leading to different statistical results so naming conventions must follow when creating the raw data .csv file. For future considerations, the system should take a database as input and have a proper interface so all naming conventions follow. An interface would allow for easier usability and allows a consumer to conduct their own experiment using the system. Lastly an accountant will be required to properly calculate taxes and inform about any other deductions.

Conclusion

To answer the questions proposed at the beginning of the experiment, selling Pokemon cards is indeed a viable way to start a business and to invest in. The statistical analysis showed incredible results showing at 30% gross margin profit and 82% rate of return with just 99 sales. With the high demand of Pokemon card products and the large influx of consumers, the results are not surprising but it is still incredible to see such high numbers. One major risk to the business I suspect is the unpredictability of the market. Are these numbers the result of the current trend? Will this market be temporary and see a huge decline in value similar to bitcoin? There is no way to completely predict the direction of the market but as of now and with the project's analysis I believe starting a business selling cards over the next few years is safe. The market does not seem to be trending downward anytime soon with more card sets releasing and more consumers joining the market everyday. Card owners should prioritize products from the set Crown Zenith as it is one of the most popular sets and analysis shows them as best selling. Single cards less than \$40 triumphs sales as the leading source of revenue.

Additional Plots of Individual Shows

Sales & Revenue Analysis — guildford_card_show

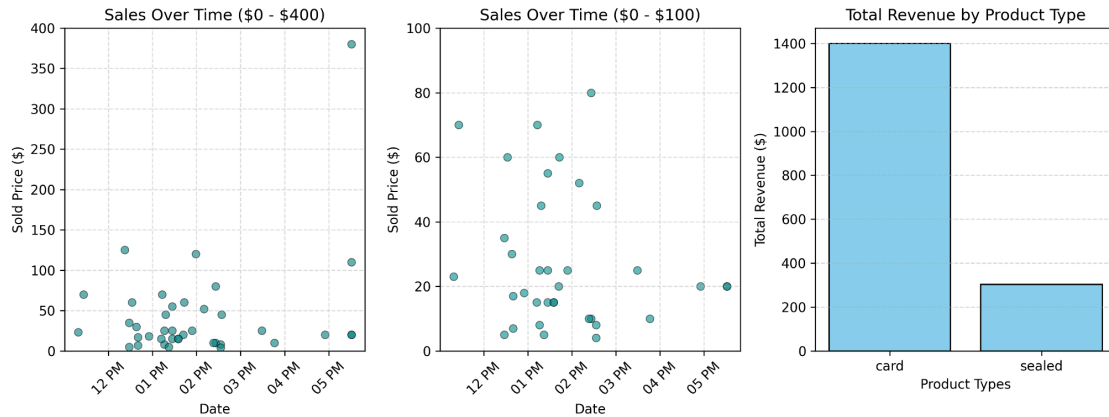


Figure 6: Sales plot of Guildford Card Show

Sales & Revenue Analysis — the_layover_show

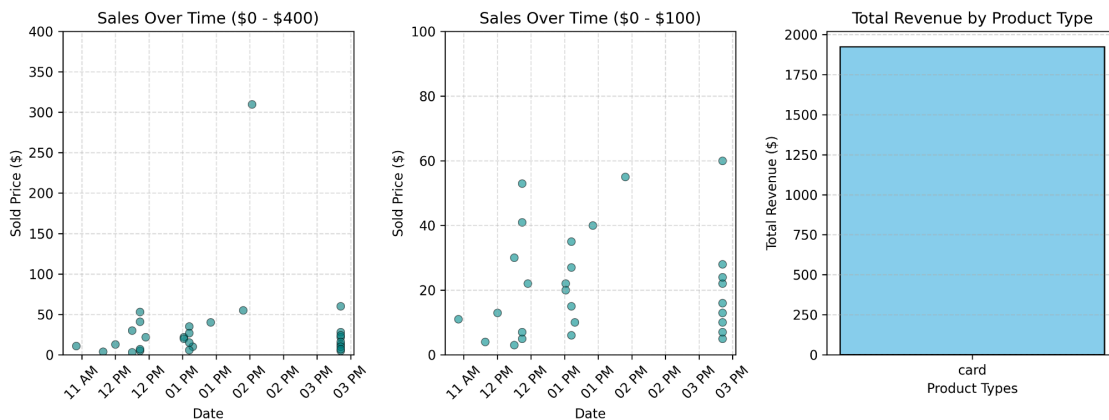


Figure 7: Sales plot of the Layover Show

Sales & Revenue Analysis — vault_card_show

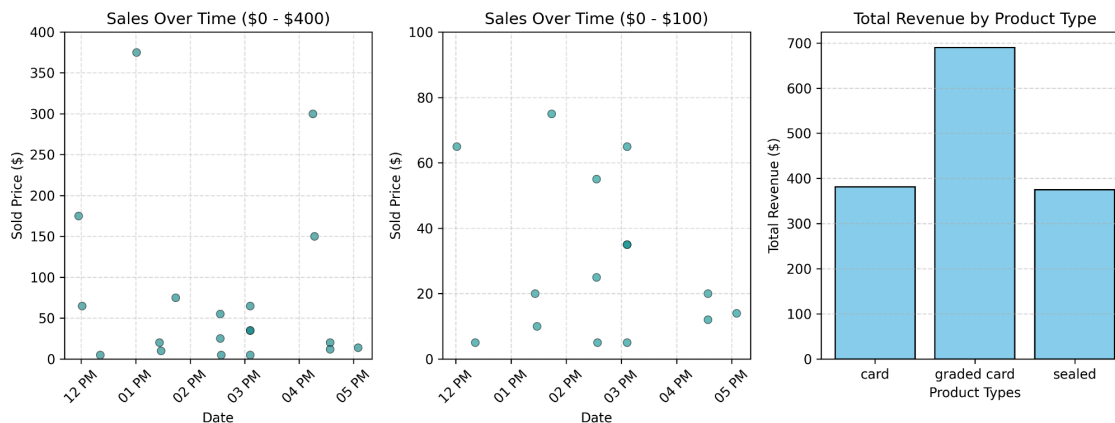


Figure 8: Sales plot of Vault Card Show

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Project Experience

Pokemon Trading Card Business Analysis Model

Simon Fraser University | Computational Data Science | April 2023 – Aug 2023

- Created a model to calculate, generate visuals and analyze business statistics using Python, Pandas, matlab and concepts from basic Data Science
- Experimented with real world sales transactions
- Cleans raw data, runs statistics and investment results and creates sales plots
- Found significant results in business potential using 99 data points from the model that suggest creating the Pokemon Trading Card business is viable
- Found results from the model that suggest better business decisions and visualizes sales over time